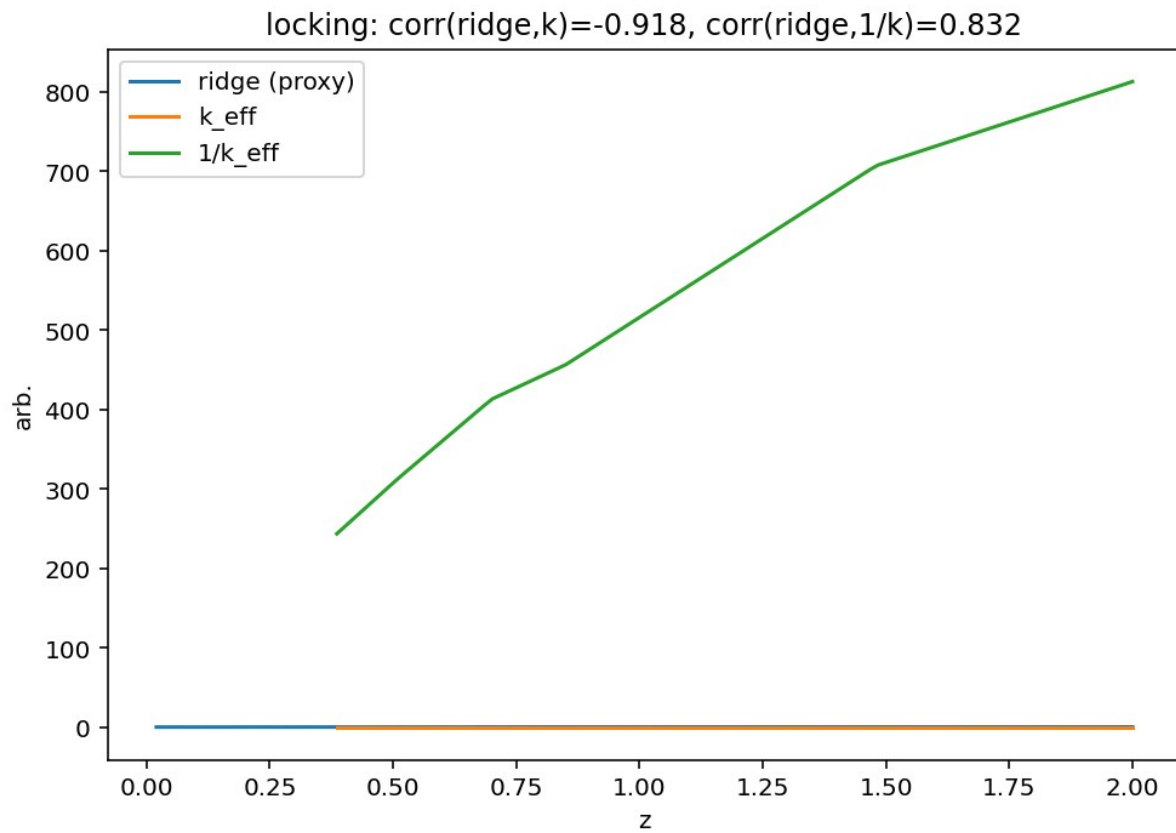


## HHT\_SN\_Residuals Results OV0



```
%run '/Users/boyde/.spyder-py3/hht_bao_lock_check.py' \
```

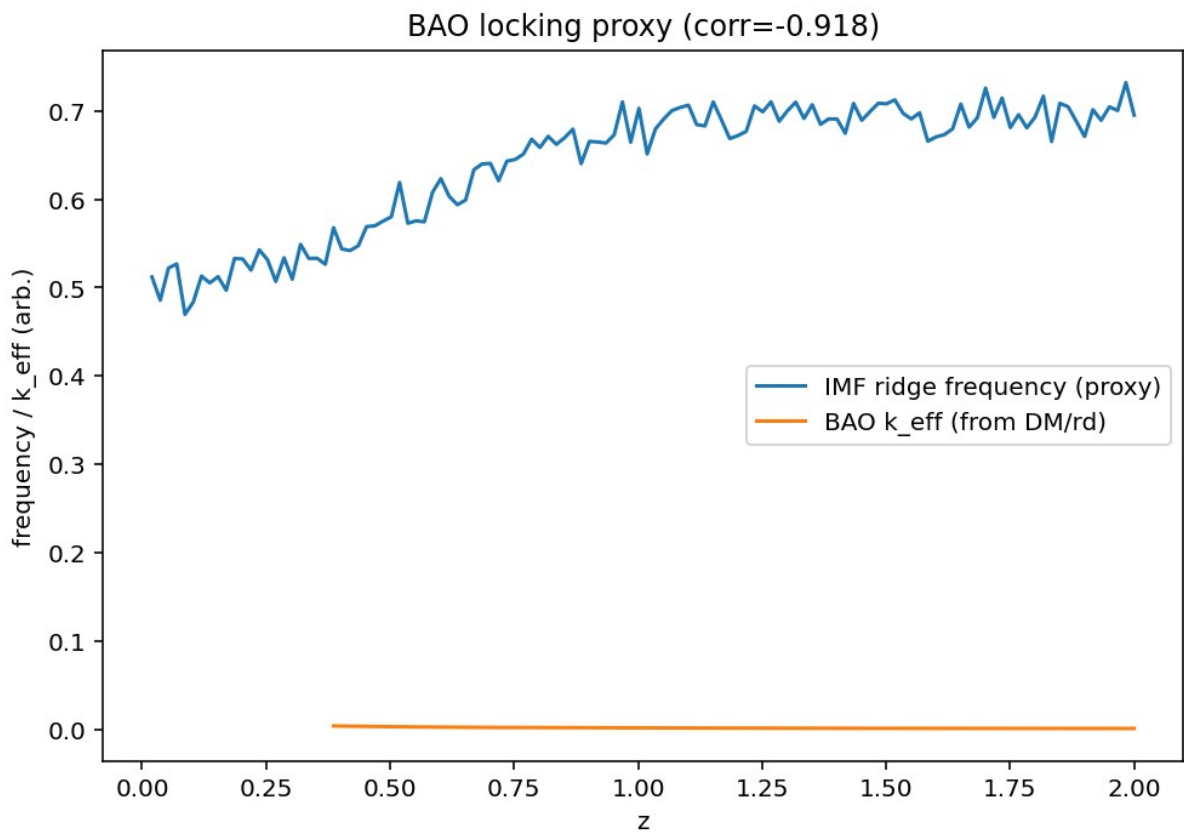
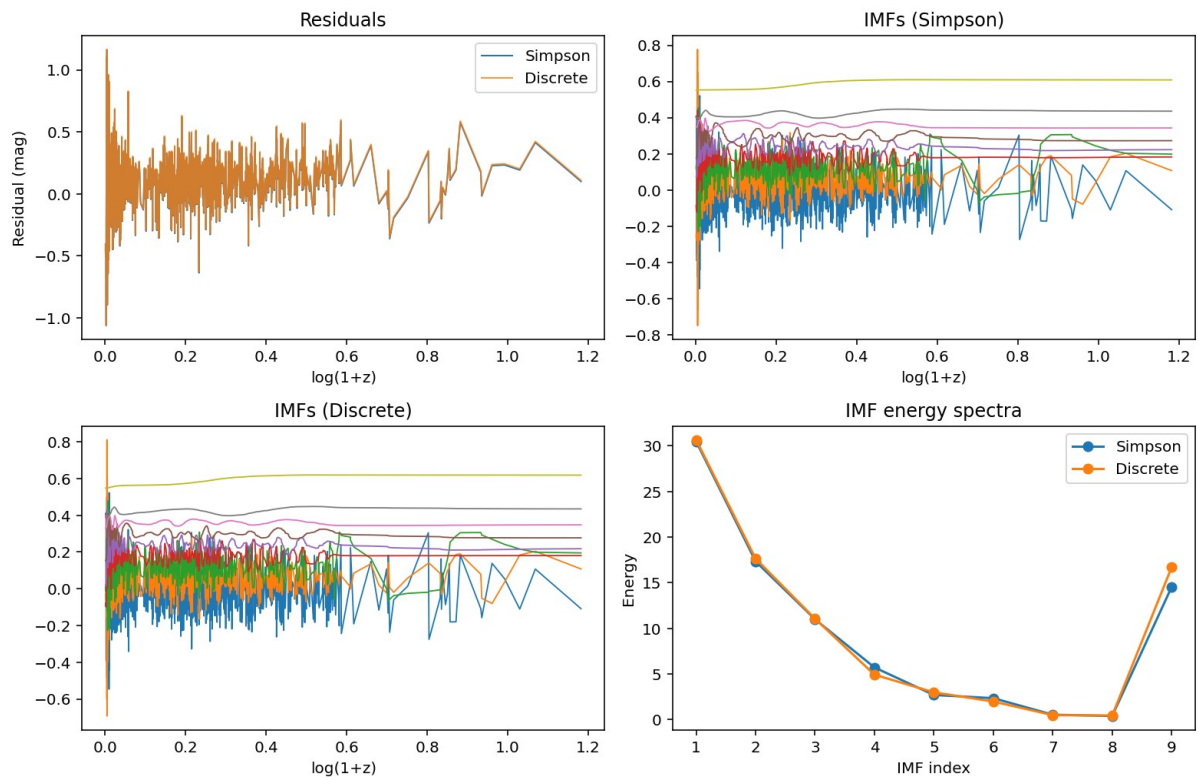
```
    '/Users/boyde/Downloads/bao_long.csv' \
```

```
    --rd 147.09 --out 'plamb_runs/hht_bao_lock_check'
```

```
[ok]  $\text{corr}(\text{ridge}, k) = -0.918$ ,  $\text{corr}(\text{ridge}, 1/k) = 0.832$ 
```

```
[ok] files at plamb_runs/hht_bao_lock_check
```

<Figure size 864x576 with 0 Axes>



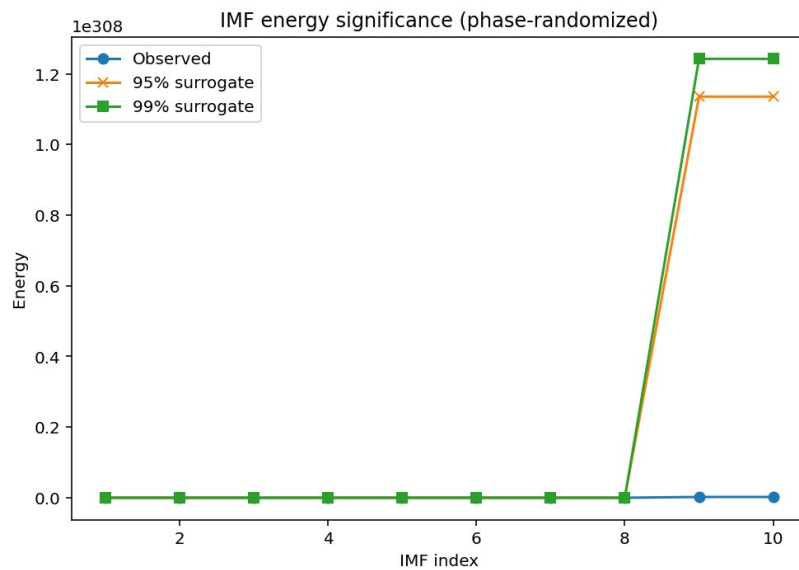
```
%run '/Users/boyde/.spyder-py3/hht_ridge_bao_lock.py' \
'/Users/boyde/Downloads/bao_long.csv' \
```

```
--rd 147.09 --out 'plamb_runs/hht_bao_lock'
```

[ok] corr = -0.918

[ok] files at plamb\_runs/hht\_bao\_lock

<Figure size 864x576 with 0 Axes>



```
%run '/Users/boyde/.spyder-py3/hht_surr_significance.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
--nsurr 500 --method simpson --out 'plamb_runs/hht_surr'
```

usage: PLamb\_Test\_10V6c\_plus.py [-h] [--out OUT] [--H0 H0] [--Om OM] [--OI OI]

[--A\_acc A\_ACC] [--n\_acc N\_ACC]

[--gamma\_c GAMMA\_C] [--epsilon\_M EPSILON\_M]

[--zmax ZMAX] [--zmin ZMIN]

[--nwalkers NWALKERS] [--nsteps NSTEPS]

[--nburn NBURN] [--seed SEED]

[--grid-points GRID\_POINTS] [--nint NINT]

[--thin THIN] [--fix\_H0] [--fix\_Om] [--fix\_OI]

[--fix\_A\_acc] [--fix\_n\_acc] [--fix\_gamma\_c]

[--fix\_epsilon\_M] [--show] [--verbose]

```

[--flat] [--save-chain] [--link-em-gc]

[--planck-c-variant]

[--model {LCDM,KdS,FR,PBH}]

[--dM-mode {linear_z,log1pz}]

[--sampler {emcee,grid,none}] [--data DATA]

[--cov COV] [--bao BAO] [--bao-cov BAO_COV]

[--cc CC] [--planck PLANCK] [--rd RD]

[--planck-zstar PLANCK_ZSTAR] [--rs RS]

[--planck-nint PLANCK_NINT]

[--kappa-em-gc KAPPA_EM_GC] [--tight-priors]

[--corner] [--diag-cov] [--preflight]

[--fluct-Lambda]

[--Lambda-model {white,OU,piecewise,spline}]

[--Lambda-amp LAMBDA_AMP]

[--Lambda-corrz LAMBDA_CORRZ]

[--Lambda-knots LAMBDA_KNOTS]

[--seed-Lambda SEED_LAMBDA]

[--couple-Lambda-Aacc]

[--lambda-aacc-gamma LAMBDA_AACC_GAMMA]

[--discrete-events] [--event-size EVENT_SIZE]

[--noether
{none,conserve_time,conserve_scale,conserve_diffeo,break_time_soft,break_scale_soft,break_diffeo_soft,break_space_soft,conserve_time_soft,conserve_scale_soft,conserve_c_soft,conserve_lum_soft,conserve_space_soft,combo_soft}]

[--noether-strength NOETHER_STRENGTH]

[--bao-c-mode {c0,c_of_z}] [--fr-nonexpanding]

[--link-sigma LINK_SIGMA] [--link-soft]

[--c-model
{linear_z,log1pz,piecewise,density_coupled,power1pz,expz,saturating,poly_log1pz,cheb_log1pz}]

[--gamma2_c GAMMA2_C] [--z_t Z_T]

[--alpha_rho ALPHA_RHO] [--c-power C_POWER]

```

```

[--c-z0 C_Z0] [--c-sat-kind {tanh,exp}]
[--c-coefs C_COEFS]
[--planck-c-mode {none,R_only,R_and_rs}]
[--rs-power RS_POWER] [--pbh-tests]
[--pbh-couple] [--pbh-M PBH_M]
[--pbh-a-dim PBH_A_DIM] [--pbh-G PBH_G]
[--pbh-c PBH_C] [--pbh-hbar PBH_HBAR]
[--pbh-kB PBH_KB] [--spec-N SPEC_N]
[--spec-beta SPEC_BETA]
[--spec-seed SPEC_SEED]
[--hht-slices HHT_SLICES] [--pbh-out PBH_OUT]
[--pbh-plots]
[--pbh-plot-slices PBH_PLOT_SLICES]

```

PLamb\_Test\_10V6c\_plus.py: error: unrecognized arguments: --residuals-only

/Users/boyde/.spyder-py3/hht\_surr\_significance.py:63: FutureWarning: The 'delim\_whitespace' keyword in pd.read\_csv is deprecated and will be removed in a future version. Use ``sep='\s+'`` instead

```
df = pd.read_csv(args.sn_table, delim_whitespace=True, comment='#')
```

/opt/anaconda3/lib/python3.12/site-packages/matplotlib/ticker.py:2119: RuntimeWarning: overflow encountered in multiply

```
steps = self._extended_steps * scale
```

[ok] CSV -> plamb\_runs/hht\_surr/hht\_imf\_significance.csv

[ok] Plot -> plamb\_runs/hht\_surr/hht\_imf\_significance.png

<Figure size 864x576 with 0 Axes>

```

%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
'plamb_runs/hht_sn_noether/combo_soft6' \
--noether combo_soft --noether-strength 6 --zmin 0.02

```

[out] writing to: /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/combo\_soft6

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$

[save] IMF stats ->

/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/combo\_soft6/hht\_sn\_imf\_stats.csv,  
/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/combo\_soft6/hht\_sn\_imf\_stats.npz

[files] /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/combo\_soft6/hht\_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
'plamb_runs/hht_sn_noether/conserv_time' \
```

```
--noether conserv_time --zmin 0.02
```

[out] writing to: /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/conserv\_time

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$

[save] IMF stats ->

/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/conserv\_time/hht\_sn\_imf\_stats.csv,  
/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/conserv\_time/hht\_sn\_imf\_stats.npz

[files] /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_noether/conserv\_time/hht\_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
'plamb_runs/hht_sn_zmin010' --zmin 0.10
```

[out] writing to: /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010/hht\_sn\_imf\_stats.csv,  
/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010/hht\_sn\_imf\_stats.npz

[files] /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010/hht\_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \  
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
'plamb_runs/hht_sn_zmin010' --zmin 0.10
```

[out] writing to: /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010/hht\_sn\_imf\_stats.csv,  
/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010/hht\_sn\_imf\_stats.npz

[files] /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin010/hht\_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \  
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
'plamb_runs/hht_sn_zmin002' --zmin 0.02
```

[out] writing to: /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin002

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin002/hht\_sn\_imf\_stats.csv,  
/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin002/hht\_sn\_imf\_stats.npz

[files] /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_zmin002/hht\_sn.png

<Figure size 864x576 with 0 Axes>

```
%run '/Users/boyde/.spyder-py3/hht_integrator_ab.py' \  
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \  
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \  
'plamb_runs/hht_sn_zmin002' --zmin 0.02
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat'
```

```
[files] plamb_runs/hht_integrator/hht_integrator_compare.png
```

```
<Figure size 864x576 with 0 Axes>
```

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

```
'/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
'plamb_runs/hht_couple/ou_seed777_ON'
```

```
[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON
```

```
[clip] kept 0.98 of points.
```

```
[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$ 
```

```
[save] IMF stats ->
```

```
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON/hht_sn_imf_stats.csv,
```

```
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON/hht_sn_imf_stats.npz
```

```
[files] /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777_ON/hht_sn.png
```

```
<Figure size 864x576 with 0 Axes>
```

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
--model FR \
```

```
--data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
--sampler none --discrete-events --event-size 0.005 \
```

```
--out 'plamb_runs/hht_compare/discrete_0005'
```

```
[status] parsing done.
```

```
[status] data loaded.
```

```
[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}
```

```
[Integrator] discrete-events enabled event-size  $\Delta z=0.005$ 
```

```
[link] soft=False kappa=1.0 sigma=0.02
```

```
Best-fit (model=FR, sampler=none): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0, 'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -615.6178439615401, 'chi2_SN': 1231.2356879230801,
```



```
'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1231.2356879230801, 'dof_total': 1694,
'dof_SN': 1694, 'chi2/dof_total': 0.7268215395059505, 'chi2_SN/dof_SN': 0.7268215395059505,
'AIC': 1245.2356879230801, 'BIC': 1283.3084890698512, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0',
'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z',
'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}
```

```
[fit] chi2_total=1231.24 dof_total=1694 chi2/dof_total=0.727 AIC=1245.24 BIC=1283.31
```

```
[fit] (SN-only) chi2_SN=1231.24 dof_SN=1694 chi2_SN/dof_SN=0.727
```

```
[timing] total 0.35 s
```

DONE 1) integrator fingerprint test (Simpson vs discrete-events)

# Simpson baseline

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
```

```
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
```

```
--sampler none --out 'plamb_runs/hht_compare/simpson'
```

```
[status] parsing done.
```

```
[status] data loaded.
```

```
[status] active params: ['H0', 'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'] frozen: {}
```

```
[link] soft=False kappa=1.0 sigma=0.02
```

```
Best-fit (model=FR, sampler=none): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A_acc': 0.0, 'n_acc': 0.0,
'gamma_c': 0.0, 'epsilon_M': 0.0, 'lnpost': -605.368998425507, 'chi2_SN': 1210.737996851014,
'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1210.737996851014, 'dof_total': 1694,
'dof_SN': 1694, 'chi2/dof_total': 0.7147213676806459, 'chi2_SN/dof_SN': 0.7147213676806459,
'AIC': 1224.737996851014, 'BIC': 1262.8107979977851, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0',
'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z',
'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}
```

```
[fit] chi2_total=1210.74 dof_total=1694 chi2/dof_total=0.715 AIC=1224.74 BIC=1262.81
```

```
[fit] (SN-only) chi2_SN=1210.74 dof_SN=1694 chi2_SN/dof_SN=0.715
```

```
[timing] total 0.04 s
```

Best-fit saved to plamb\_runs/hht\_compare/simpson/FR\_none\_bestfit.txt

DONE 1) integrator fingerprint test (Simpson vs discrete-events)

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
```

'/Users/boyde/.spyder-py3/PLamb\_Test\_10V6c\_plus.py' \

'/Users/boyde/Downloads/Pantheon+SH0ES.dat' \

'plamb\_runs/hht\_sn\_FRPBH'

[out] writing to: /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_FRPBH

[clip] kept 0.98 of points.

[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$

[save] IMF stats -> /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_FRPBH/hht\_sn\_imf\_stats.csv,  
/Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_FRPBH/hht\_sn\_imf\_stats.npz

[files] /Users/boyde/.spyder-py3/plamb\_runs/hht\_sn\_FRPBH/hht\_sn.png

<Figure size 864x576 with 0 Axes>

[status] parsing done.

[status] data loaded.

[status] active params: ['H0', 'Om', 'Ol', 'A\_acc', 'n\_acc', 'gamma\_c', 'epsilon\_M'] frozen: {}

[link] soft=False kappa=1.0 sigma=0.02

Best-fit (model=FR, sampler=None): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A\_acc': 0.0, 'n\_acc': 0.0, 'gamma\_c': 0.0, 'epsilon\_M': 0.0, 'lnpost': -605.368998425507, 'chi2\_SN': 1210.737996851014, 'chi2\_BAO': 0.0, 'chi2\_CC': 0.0, 'chi2\_Planck': 0.0, 'chi2': 1210.737996851014, 'dof\_total': 1694, 'dof\_SN': 1694, 'chi2/dof\_total': 0.7147213676806459, 'chi2\_SN/dof\_SN': 0.7147213676806459, 'AIC': 1224.737996851014, 'BIC': 1262.8107979977851, 'N\_SN': 1701, 'N\_eff': 1701, 'active': ['H0', 'Om', 'Ol', 'A\_acc', 'n\_acc', 'gamma\_c', 'epsilon\_M'], 'frozen': {}, 'w\_eff': -1.0, 'bao\_c\_mode': 'c\_of\_z', 'c\_model': 'linear\_z', 'planck\_c\_mode': 'none', 'rs\_power': 1.0}

[fit] chi2\_total=1210.74 dof\_total=1694 chi2/dof\_total=0.715 AIC=1224.74 BIC=1262.81

[fit] (SN-only) chi2\_SN=1210.74 dof\_SN=1694 chi2\_SN/dof\_SN=0.715

[timing] total 0.04 s

Best-fit saved to plamb\_runs/hht\_compare/simpson/FR\_none\_bestfit.txt

DONE # discrete-events ( $\Delta z = 0.01$ )

%run '/Users/boyde/.spyder-py3/PLamb\_Test\_10V6c\_plus.py' \

--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \

--sampler none --discrete-events --event-size 0.01 \

--out 'plamb\_runs/hht\_compare/discrete\_001'

[status] parsing done.

[status] data loaded.

[status] active params: ['H0', 'Om', 'Ol', 'A\_acc', 'n\_acc', 'gamma\_c', 'epsilon\_M'] frozen: {}

[Integrator] discrete-events enabled event-size  $\Delta z=0.01$

[link] soft=False kappa=1.0 sigma=0.02

Best-fit (model=FR, sampler=None): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A\_acc': 0.0, 'n\_acc': 0.0, 'gamma\_c': 0.0, 'epsilon\_M': 0.0, 'lnpost': -625.5295460871644, 'chi2\_SN': 1251.0590921743287, 'chi2\_BAO': 0.0, 'chi2\_CC': 0.0, 'chi2\_Planck': 0.0, 'chi2': 1251.0590921743287, 'dof\_total': 1694, 'dof\_SN': 1694, 'chi2/dof\_total': 0.738523667163122, 'chi2\_SN/dof\_SN': 0.738523667163122, 'AIC': 1265.0590921743287, 'BIC': 1303.1318933210998, 'N\_SN': 1701, 'N\_eff': 1701, 'active': ['H0', 'Om', 'Ol', 'A\_acc', 'n\_acc', 'gamma\_c', 'epsilon\_M'], 'frozen': {}, 'w\_eff': -1.0, 'bao\_c\_mode': 'c\_of\_z', 'c\_model': 'linear\_z', 'planck\_c\_mode': 'none', 'rs\_power': 1.0}

[fit] chi2\_total=1251.06 dof\_total=1694 chi2/dof\_total=0.739 AIC=1265.06 BIC=1303.13

[fit] (SN-only) chi2\_SN=1251.06 dof\_SN=1694 chi2\_SN/dof\_SN=0.739

[timing] total 0.18 s

Best-fit saved to plamb\_runs/hht\_compare/discrete\_001/FR\_none\_bestfit.txt

DONE # realize  $\Lambda(z)$  once (OU) and keep params fixed (sampler none)

```
%run '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
--model FR --data '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
--sampler none --fluct-Lambda --Lambda-model OU --Lambda-amp 1e-120 \
--Lambda-corrz 0.3 --seed-Lambda 777 \
--out 'plamb_runs/hht_couple/ou_seed777'
```

[status] parsing done.

[status] data loaded.

[status] active params: ['H0', 'Om', 'Ol', 'A\_acc', 'n\_acc', 'gamma\_c', 'epsilon\_M'] frozen: {}

[Lambda] model=OU amp=1.000e-120 corrz=0.3 knots=6 seed=777 coupling=False gamma=0.0

[link] soft=False kappa=1.0 sigma=0.02

Best-fit (model=FR, sampler=None): {'H0': 70.0, 'Om': 0.3, 'Ol': 0.7, 'Ok': 0.0, 'A\_acc': 0.0, 'n\_acc': 0.0, 'gamma\_c': 0.0, 'epsilon\_M': 0.0, 'lnpost': -605.368998425507, 'chi2\_SN': 1210.737996851014,

```
'chi2_BAO': 0.0, 'chi2_CC': 0.0, 'chi2_Planck': 0.0, 'chi2': 1210.737996851014, 'dof_total': 1694,
'dof_SN': 1694, 'chi2/dof_total': 0.7147213676806459, 'chi2_SN/dof_SN': 0.7147213676806459,
'AIC': 1224.737996851014, 'BIC': 1262.8107979977851, 'N_SN': 1701, 'N_eff': 1701, 'active': ['H0',
'Om', 'Ol', 'A_acc', 'n_acc', 'gamma_c', 'epsilon_M'], 'frozen': {}, 'w_eff': -1.0, 'bao_c_mode': 'c_of_z',
'c_model': 'linear_z', 'planck_c_mode': 'none', 'rs_power': 1.0}
```

```
[fit] chi2_total=1210.74 dof_total=1694 chi2/dof_total=0.715 AIC=1224.74 BIC=1262.81
```

```
[fit] (SN-only) chi2_SN=1210.74 dof_SN=1694 chi2_SN/dof_SN=0.715
```

```
[timing] total 0.04 s
```

```
Best-fit saved to plamb_runs/hht_couple/ou_seed777/FR_none_bestfit.txt
```

DONE 2)  $\Lambda(z)$  coupling / phase-locking probe

```
%run '/Users/boyde/.spyder-py3/hht_sn_residuals.py' \
      '/Users/boyde/.spyder-py3/PLamb_Test_10V6c_plus.py' \
      '/Users/boyde/Downloads/Pantheon+SH0ES.dat' \
      'plamb_runs/hht_couple/ou_seed777'
```

```
[out] writing to: /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777
```

```
[clip] kept 0.98 of points.
```

```
[grid] uniform L bins = 1111,  $L \in [0.0012, 1.1821]$ 
```

```
[save] IMF stats ->
```

```
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777/hht_sn_imf_stats.csv,
/Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777/hht_sn_imf_stats.npz
```

```
[files] /Users/boyde/.spyder-py3/plamb_runs/hht_couple/ou_seed777/hht_sn.png
```

```
<Figure size 864x576 with 0 Axes>
```